

- (8) **1.** Solve each system (by reducing its augmented matrix to RREF) or show that it is inconsistent. If it has infinitely many solutions, express these parametrically.

$$(a) \begin{cases} 2x + 3y + 4z = -1 \\ x + 3y + 5z = 1 \\ -4x + 2y + 6z = 9 \end{cases}$$

$$(b) \begin{cases} 2x_1 + 6x_2 + x_3 = 17 \\ 3x_1 + 9x_2 + x_3 - 2x_4 = 22 \end{cases}$$

- (5) **2.** At the Chicken Nugget Place, boxes of nuggets are sold in three sizes: a small box contains 3 nuggets, a medium box contains 5 nuggets, and a large box contains 8 nuggets. One day 7 boxes and a total of 41 nuggets are sold. You would like to know how many of each size box were sold.

- (a) Define all the necessary variables (using the order provided) and set up the system of equations to answer the question.

Do not solve.

- (b) Given the reduced row echelon matrix to the system is $\left[\begin{array}{ccc|c} 1 & 0 & -3/2 & -3 \\ 0 & 1 & 5/2 & 10 \end{array} \right]$, give the general solution to the system.

- (c) Find all possible realistic solutions.

- (4) **3.** Find all values of h such that the augmented matrix

$$\left[\begin{array}{ccc|c} h & (h-1) & (h-1) & (h-1) \\ 0 & (h-2) & (h-2) & (h-2) \\ 0 & 0 & (h-3) & (h-4) \end{array} \right]$$

represents a system with:

- (a) a unique solution. (b) infinitely many solutions. (c) no solution.

- (3) **4.** Suppose A is 3×4 , B is 4×4 and C is 4×5 .

Determine the size, $m \times n$, of each of the following matrices, or state that the expression is “undefined”.

- (a) $A(BC)$ (b) C^2 (c) $A^T A - B^3$

- (3) **5.** Given the matrices $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & k \\ 0 & 5 \end{bmatrix}$, find all the value(s) of k (if any) so that $AB = BA$.

- (4) **6.** Let A and B be $n \times n$ matrices. Solve for X in the matrix equation $X^T = (AX)^T + B$.

(All necessary matrices are assumed to be invertible.)

- (6) **7.** Consider the matrix $A = \begin{bmatrix} 2 & -3 & 1 \\ 0 & 1 & 5 \\ 1 & 0 & -4 \end{bmatrix}$.

- (a) Find $\det(A)$. (b) Find $\text{adj}(A)$. (c) Use this information to find A^{-1} .

- (4) **8.** Suppose that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 4$.

Evaluate each of the following or state that there is not enough information to do so.

$$(a) \begin{vmatrix} d & e & f \\ 2a & 2b & 2c \\ 5g - 7d & 5h - 7e & 5i - 7f \end{vmatrix}$$

$$(b) \begin{vmatrix} -a & 8d & a + d \\ -b & 8e & b + e \\ -c & 8f & c + f \end{vmatrix}$$

- (4) **9.** Suppose A, B and C are 4×4 matrices with $\det(A) = 10$, $\det(B) = -3$ and C is not invertible. Evaluate each of the following or state that there is not enough information to do so.
- (a) $\det[(2A)^{-1}]$ (b) $\det(C^T B)$ (c) $\det(AB + C)$
- (8) **10.** Given the points $P_1(1, 3, 2)$, $P_2(3, -1, 4)$, $P_3(3, 5, -2)$ and $P_4(1, 3, -2)$
- (a) Find a unit vector in the opposite direction of $\overrightarrow{P_1 P_2}$
- (b) Find a **vector** equation of the line which passes through points P_2 and P_3
- (c) Find a **general (standard)** equation of the plane with points P_1, P_2 and P_3
- (d) Is P_4 on the plane?
- (3) **11.** Given the planes $3x + 6y - 9z = 3$ and $4x + 8y - 12z = 4$,
- (a) Write a vector equation of their intersection.
- (b) Is the intersection found in part (a) a point, a line, or a plane?
- (2) **12.** Given $\mathcal{P}_1 : 7x - 12y + 10z = 2$, $\mathcal{P}_2 : 6x - 9y + kz = 7$, find the value of k that makes $\mathcal{P}_1$ and $\mathcal{P}_2$ perpendicular, if possible.
- (7) **13.** *Winter Warrior*, a Canadian winter boot manufacturer, has four factories whose daily production (in 100's) of children's, women's, and men's boots is shown in the table:

	Factory 1	Factory 2	Factory 3	Factory 4
Children's	2	6	1	4
Women's	1	5	2	3
Men's	2	7	2	4

Winter Warrior has a contract with a large retail chain which requires a weekly production of 5000 children's boots, 3900 women's boots, and 5600 men's boots. That demand is currently met by 6 days of operation at Factory 1, 4 days at Factory 2, 2 days at Factory 3, and 3 days at Factory 4.

- (a) Using the production vectors $\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4$, and demand vector $\vec{v}$, give the linear combination that represents the current production schedule.
- (b) Use the reduction provided below to find the general solution for all theoretical production schedules.
- $$\left[\begin{array}{cccc|c} 2 & 6 & 1 & 4 & 50 \\ 1 & 5 & 2 & 3 & 39 \\ 2 & 7 & 2 & 4 & 56 \end{array} \right] \text{ reduces to: } \left[\begin{array}{cccc|c} 1 & 0 & 0 & -3 & -3 \\ 0 & 1 & 0 & 2 & 10 \\ 0 & 0 & 1 & -2 & -4 \end{array} \right]$$
- (c) *Winter Warrior* would like to close one of the factories and shift production to the other three. Using your general solution from part (b), determine which of the factory closures are feasible and for each of these, give the resulting production schedule.

- (7) **14.** A college student alternates between two evening routines: studying or relaxing. Based on past behaviour, if the student studies on a given evening, then the next evening there is a probability of 0.7 that they will study again and a probability of 0.3 that they will relax. If the student relaxes on a given evening, then the next evening there is a probability of 0.4 that they will study and a probability of 0.6 that they will relax again.
- (a) Construct a transition matrix for this situation.
 - (b) If the student studies tonight, give the initial state vector and then determine the probability that they will be relaxing two nights from now.
 - (c) Find the steady-state vector $\vec{q}$ for this situation. Give your final answer using simplified fractions.
 - (d) Interpret your result from part (c) by determining the long-run percentage of evenings the student spends studying.
- (5) **15.** On a university campus, there are two main student-run services: Food Services and the Campus Store. For each dollar of goods produced by Food Services, it takes \$0.20 of its own output and \$0.40 of Campus Store's output. Each dollar of output from the Campus Store uses \$0.10 of Food Services output and \$0.20 of its own output.
- (a) Find a consumption matrix associated with this economy.
 - (b) What production will satisfy a demand for \$1200 in Food Services and \$1800 from the Campus Store?
 - (c) In the above scenario what is the internal consumption of each service?
- (6) **16.** An environmental system consists of three sectors: forests, agriculture, and water resources. Each sector depends on the others to sustain its activity, and since all output is used between the sectors, it is a closed model. To sustain one dollar of forest output, the system requires \$0.40 from forests, \$0.20 from agriculture, and \$0.40 from water resources. For each dollar of agricultural output, it requires \$0.30 from forests, \$0.40 from agriculture, and \$0.30 from water resources. Each dollar of water resource output requires \$0.30 from forests, \$0.30 from agriculture, and \$0.40 from water resources.
- (a) Find an exchange/consumption matrix associated with this system.
 - (b) If the forest output is worth \$1 800, how much should the other two sectors produce?
- (5) **17.** Minimize $z = -5x_1 + 2x_2 - 7x_3$ subject to the constraints:

$$\begin{cases} -3x_1 + 2x_2 + x_3 \leq 12 \\ 4x_1 - 2x_2 + x_3 \leq 8 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

- (5) **18.** Consider the optimization problem: Maximize $z = 2x_1 - 4x_2 + 5x_3$ subject to the constraints:

$$\begin{cases} 3x_1 + 5x_2 - 2x_3 \leq 20 \\ -6x_1 + 2x_2 + 3x_3 \leq 9 \\ 2x_1 - 5x_2 - 2x_3 \leq 5 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

- (a) Use the Simplex Method to show that the linear program is unbounded (no solution).
- (b) Find a feasible solution for which $z \geq 12015$.

- (5) **19.** An airline company sells 3 different kinds of plane tickets: Economy Class for \$210, Business Class for \$405 and First Class for \$465. The costs to accommodate each passenger is \$50 for Economy Class, \$150 for Business Class and \$150 for First Class. The space allocated for luggage is 1m^3 for Economy Class, 2m^3 for Business Class and 3m^3 for First Class.

The company has a total budget of \$26 000 to operate the flight, can use up to 300m^3 for luggage and can fit a maximum of 220 seats on the plane. They would like to know how many tickets of each class they should sell in order to maximize their revenue.

- (a) Define all the necessary variables (using the order provided). State the objective and the objective function. List all the constraints. **Do not solve.**
- (b) Using the final Simplex table below, determine the maximum revenue and how many tickets of each class should be sold.

$$\left[\begin{array}{c|cccc|cccc} 1 & 0 & 0 & 135 & 0 & 195 & 15 & 61800 \\ \hline 0 & 0 & 0 & -100 & 1 & -100 & 50 & 7000 \\ 0 & 0 & 1 & 2 & 0 & 1 & -1 & 80 \\ 0 & 1 & 0 & -1 & 0 & -1 & 2 & 140 \end{array} \right]$$

A B C	D E F	G H I	J K L	M N O	P Q R	S T U	V W X	Y Z
1 2 3	4 5 6	7 8 9	10 11 12	13 14 15	16 17 18	19 20 21	22 23 24	25 26

a	1	3	5	7	9	11	15	17	19	21	23	25
a^{-1}	1	9	21	15	3	19	7	23	11	5	17	25

- (6) **20.** The end of a sentence was encrypted using the matrix $A = \begin{bmatrix} 5 & 3 \\ 17 & 8 \end{bmatrix}$.

- (a) Find a decoding matrix A^{-1} .
- (b) Verify your answer in **a)** by calculating $A \cdot A^{-1}$
- (c) Decrypt the end of the sentence: Any fool can criticize, condemn and complain – and **FCYOMF**.

ANSWERS:

1. (a) $(-3/2, 0, 1/2)$

(b) $(5 - 3s + 2t, s, 7 - 4t, t)$

$$2. \quad \begin{array}{l} x_1 = \text{\#of sets of 3 nuggets} \\ x_2 = \text{\#of sets of 5 nuggets} \\ x_3 = \text{\#of sets of 8 nuggets} \end{array} \quad \left\{ \begin{array}{l} x_1 + x_2 + x_3 = 7 \\ 3x_1 + 5x_2 + 8x_3 = 41 \end{array} \right.$$

(b) $(-3 + 3t/2, 10 - 5t/2, t)$

(c) $t = 2 \rightarrow (0, 5, 2), t = 4 \rightarrow (3, 0, 4)$

3. (a) $h \neq 0, 2, 3$

(b) $h = 0, h = 2$

(c) $h = 3$

4. (a) 3×5

(b) undefined

(c) 4×4

5. $k = 1$

6. $X = (I - A)^{-1}B^T$

7. (a) $\det(A) = -24$

$$(b) \operatorname{adj}(A) = \begin{bmatrix} -4 & -12 & -16 \\ 5 & -9 & -10 \\ -1 & -3 & 2 \end{bmatrix} \quad (c) A^{-1} = \frac{1}{24} \begin{bmatrix} 4 & 12 & 16 \\ -5 & 9 & 10 \\ 1 & 3 & -2 \end{bmatrix}$$

8. (a) -40

(b) 0

9. (a) $1/160$

(b) 0

(c) Not enough information

$$10. \quad (a) \frac{1}{\sqrt{24}} \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix} \quad (b) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} + t \begin{bmatrix} 0 \\ 6 \\ -6 \end{bmatrix} \quad \begin{array}{l} (c) x + y + z = 6 \\ (d) \text{No} \end{array}$$

$$11. \quad (a) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

(b) A plane

12. $k = -15$

13. (a) $6\vec{u}_1 + 4\vec{u}_2 + 2\vec{u}_3 + 3\vec{u}_4 = \vec{v}$

(b) $c_1 = -3 + 3t, c_2 = 10 - 2t, c_3 = -4 + 2t, c_4 = t$

(c) Close Factory 3: $3\vec{u}_1 + 6\vec{u}_2 + 2\vec{u}_4 = \vec{v}$

$$14. \quad (a) P = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \quad (b) x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad 39\% \quad (c) q = \begin{bmatrix} 4/7 \\ 3/7 \end{bmatrix} \quad (d) 4/7 \approx 57.14\%$$

$$15. \quad (a) C = \begin{bmatrix} 0.2 & 0.1 \\ 0.4 & 0.2 \end{bmatrix}. \quad (b) X = \begin{bmatrix} 1900 \\ 3200 \end{bmatrix} \quad (c) X - C = \begin{bmatrix} 700 \\ 1400 \end{bmatrix}$$

$$16. \quad (a) E = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.3 \\ 0.4 & 0.3 & 0.4 \end{bmatrix}$$

(b) \$1600 agriculture and \$2000 water

17. $z = -68$ at $(0, 1, 10, 0, 0)$

18. (a) From the table $\left[\begin{array}{c|ccc|ccc} 1 & -12 & 22/3 & 0 & 0 & 5/3 & 0 & 15 \\ \hline 0 & -1 & 19/3 & 0 & 1 & 2/3 & 0 & 26 \\ 0 & -2 & 2/3 & 1 & 0 & 1/3 & 0 & 3 \\ 0 & -2 & -11/3 & 0 & 0 & 2/3 & 1 & 11 \end{array} \right]$ we see that x_1 is unbounded.

(b) $z = 12\,015$ at $(1000, 0, 2003, 1026, 0, 2011)$ - Many other possible answers.

19. (a) $x_1 = \#$ of tickets in Economy Class $x_2 = \#$ of tickets in Business Class $x_3 = \#$ of tickets in First Class,
Maximize $R = 210x_1 + 405x_2 + 465x_3$ subject to:

$$\begin{cases} 50x_1 + 150x_2 + 150x_3 \leq 26\,000 \\ x_1 + 2x_2 + 3x_3 \leq 300 \\ x_1 + x_2 + x_3 \leq 220 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

(b) $R = 61\,800$ when

$x_1 = 140$ Economy Class tickets, $x_2 = 80$ Business Class tickets and $x_3 = 0$ First Class tickets.

20. (a) $A^{-1} = \begin{bmatrix} 4 & 5 \\ 11 & 9 \end{bmatrix}$

(b) $A \cdot A^{-1} = \begin{bmatrix} 53 & 52 \\ 156 & 157 \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) MOST DO